

SYSTEM SECURITY AUDIT REPORT

Lynis Security Assessment

Report Generated: 2026-03-13 15:15:59

System: guacamole

IP Address: 192.168.1.101

OS: Ubuntu 24.04

Kernel: 6.8.0-88-generic

Owner: Security Team

Audit Standard: ISO 27001 / NIST / CIS Benchmarks

This report contains confidential security audit information. Unauthorized access, use, or distribution is prohibited. This assessment is based on system configuration at the time of audit.

1. EXECUTIVE SUMMARY

Metric	Value	Status
Overall Security Score	95/100	Score
Risk Level	Low	Risk
Critical Issues	1	Warnings
Recommendations	50	Suggestions
Audit Scope	Full System Assessment	Scope

This security audit report provides a comprehensive assessment of the system's current security posture. The system achieved an overall compliance score of **95/100**, indicating a **Low** risk level. The assessment identified 1 critical issue(s) requiring immediate attention and 50 recommendations for security improvement.

The audit was conducted using Lynis, a comprehensive security auditing tool that evaluates system configuration, security controls, and compliance against industry standards including ISO 27001, NIST, and CIS Benchmarks.

2. SYSTEM OVERVIEW

System Identification:

Hostname: guacamole

IP Address: 192.168.1.101

Owner: Security Team

Operating System:

OS: Ubuntu 24.04

Kernel Version: Unknown

Hardware Platform: Unknown

Virtual Machine: 1

Environment:

System Boot Time (days): 107

Service Manager: systemd

CPU Capabilities: PAE Enabled, NX Enabled

3. CRITICAL FINDINGS

1. KRNL-5830

Reboot of system is most likely needed

Recommendation: text:reboot

4. SECURITY RECOMMENDATIONS

Total Recommendations: 50 (Critical: 0, High: 16, Medium: 8, Low: 26)

High Priority

1. AUTH-9229

Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values
Solution:

2. AUTH-9230

Configure password hashing rounds in /etc/login.defs
Solution:

3. AUTH-9262

Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc
Solution:

4. AUTH-9282

When possible set expire dates for all password protected accounts
Solution:

5. AUTH-9286

Configure minimum password age in /etc/login.defs
Solution:

... and 11 more high priority items

Medium Priority

1. FILE-6310

To decrease the impact of a full /home file system, place /home on a separate partition
Solution:

2. FILE-6310

To decrease the impact of a full /tmp file system, place /tmp on a separate partition
Solution:

3. FILE-6310

To decrease the impact of a full /var file system, place /var on a separate partition
Solution:

4. NETW-3200

Determine if protocol 'dccp' is really needed on this system
Solution:

5. NETW-3200

Determine if protocol 'sctp' is really needed on this system
Solution:

... and 3 more medium priority items

Low Priority

1. LYNIS

This release is more than 4 months old. Check the website or GitHub to see if there is an update available.
Solution:

2. DEB-0280

Install libpam-tmpdir to set \$TMP and \$TMPDIR for PAM sessions
Solution:

3. DEB-0810

Install apt-listbugs to display a list of critical bugs prior to each APT installation.

Solution:

4. DEB-0811

Install apt-listchanges to display any significant changes prior to any upgrade via APT.

Solution:

5. DEB-0880

Install fail2ban to automatically ban hosts that commit multiple authentication errors.

Solution:

... and 21 more low priority items

5. SECURITY CONTROLS ASSESSMENT

Control Area	Status	Notes
Authentication & Access Control	Enabled	PAM modules configured
Firewall	Not Found	iptables rules in place
SSH Hardening	Enabled	Review SSH configuration
File Integrity Monitoring	Not Installed	Consider AIDE/Tripwire
System Auditing	Enabled	Enable auditd for logging
Kernel Hardening	Partial	Review sysctl settings

6. COMPLIANCE & STANDARDS

Assessment Framework:

This audit evaluates system compliance against the following standards:

- **ISO/IEC 27001** - Information Security Management System
- **NIST Cybersecurity Framework** - Identify, Protect, Detect, Respond, Recover
- **CIS Benchmarks** - Center for Internet Security Best Practices
- **SANS Top 25** - Most Dangerous Software Errors

Control Domains Evaluated:

- ✓ System Initialization & Boot Security
- ✓ File Systems & Storage Protection
- ✓ Access Control & Authentication
- ✓ Communication & Network Security
- ✓ System & Software Maintenance
- ✓ Security Tools & Monitoring
- ✓ Cryptography & Encryption
- ✓ Logging & Auditing

7. REMEDIATION ACTION PLAN

Immediate Actions (0-7 days):

1. Address all critical findings listed in Section 3
2. Review and harden SSH configuration
3. Enable system auditing and file integrity monitoring

Short-term Actions (1-4 weeks):

1. Implement high-priority recommendations from Section 4
2. Deploy additional security monitoring tools
3. Update all system packages and apply security patches

Long-term Actions (1-3 months):

1. Implement medium and low priority recommendations
2. Establish continuous monitoring and regular audit schedule
3. Re-run audit to verify remediation effectiveness

Ongoing Maintenance:

- Schedule monthly security audits
- Review audit logs regularly
- Keep audit tools and system software updated

8. CONCLUSION

The security audit of **guacamole** reveals a current security posture with a compliance score of **95/100**, indicating a **Low** risk level.

While the system demonstrates several security controls in place, there are areas requiring immediate attention to improve the overall security posture. Implementation of the recommended actions will significantly enhance the system's ability to protect against security threats.

Next Steps:

1. Distribute this report to authorized personnel
2. Establish a remediation timeline based on risk levels
3. Assign responsibility for implementing recommendations
4. Schedule a follow-up audit in 30-60 days to verify improvements

This report should be reviewed regularly and updated with each new audit cycle.